

Category Boundary Sorting Teacher Notes

TOK Cognitive Science Activity Bank • Knowledge and Language • 25-35 min

Category	Knowledge and Language	Time	25-35 min
Difficulty	Medium	ID	category-boundary-sorting

Big idea: Categories are made for purposes.

Overview

Students investigate how categories depend on criteria and purposes.

Student Prompt

Inspect Category Boundary Sorting Stimulus A. Record one first claim, the evidence you used, your confidence, and what would make you revise.

Concrete Stimulus Packet

Category Boundary Sorting Stimulus A	Student-facing stimulus 1: Cards such as game, sport, art, science, medicine, belief, evidence, and opinion. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Students inspect this exact card first and commit to a claim before hearing anyone else.
Category Boundary Sorting Stimulus B	Student-facing stimulus 2: Borderline cases that force students to define category rules. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Use this for comparison, a second round, or a partner challenge.
Category Boundary Sorting Reveal / Twist	Student-facing stimulus 3: A rule revision round after students encounter exceptions. Annotate the wording. Circle the words that change tone, emotion, implication, or trust.	Teacher reveal: connect the changed response to the big idea — Categories are made for purposes.

Actual Lesson Questions and Key

#	Question for students	Teacher key / cue
1	After inspecting Category Boundary Sorting Stimulus A, what is your first knowledge claim?	Teacher cue: look for a clear claim, not just a description.
2	Which exact detail from Category Boundary Sorting Stimulus A did you use as evidence?	Teacher cue: students should name visible words, data, source features, method features, or contextual clues.
3	What did you infer beyond the evidence?	Teacher cue: this is where assumptions, framing, models, or perspective usually appear.
4	How confident are you: 50%, 60%, 70%, 80%, 90%, or 100%?	Teacher cue: confidence is data for the debrief, not a grade.
5	What missing information would most improve the knowledge claim?	Teacher cue: push for method, source, context, comparison, or definitions.
6	Do categories describe reality or organise it?	Teacher cue: students should move from the classroom example to a general TOK issue.

Teacher Key / Reveal

The key move is not the “right answer”; it is the moment students see how categories are made for purposes.

- Strong response: separates observation from inference.
- Strong response: names a method, source, model, language choice, context, or perspective.
- Strong response: revises confidence when new evidence or a new criterion appears.
- Weak response: gives only opinion or says “it depends” without criteria.

Setup Checklist

- Prepare a starter stimulus using: Cards such as game, sport, art, science, medicine, belief, evidence, and opinion.
- Print or project the student response grid before the activity begins.
- Plan a private first-response moment before any group discussion.

Ready-to-Use Examples

- Cards such as game, sport, art, science, medicine, belief, evidence, and opinion.
- Borderline cases that force students to define category rules.
- A rule revision round after students encounter exceptions.

Suggested Timing

Stage	Teacher move	Watch for
1	Groups sort prompts into yes, no and depends.	Assumptions, confidence shifts, evidence claims, and language choices.
2	For each "depends", they must state the criterion.	Assumptions, confidence shifts, evidence claims, and language choices.
3	Groups trade criteria and test them on borderline cases.	Assumptions, confidence shifts, evidence claims, and language choices.
4	Discuss why different communities define the same term differently.	Assumptions, confidence shifts, evidence claims, and language choices.
Debrief	Move from the activity result to a TOK knowledge question.	Students distinguishing method, evidence, interpretation, and overclaiming.

Teacher Language

- Write your first judgement before we discuss it; the first answer is useful evidence.
- Separate what you noticed from what you inferred.
- What would make this knowledge claim more reliable?

Common Misconceptions

- Students may say language merely expresses thought rather than shaping interpretation.
- Students may identify bias in extreme examples but miss subtle framing.

Assessment Look-Fors

- Explains how wording changes interpretation before evidence is judged.
- Uses concrete language choices as evidence for a TOK claim.

Differentiation and Adaptations

- Short lesson: use only the first example, one response grid, and one debrief question.
- Long lesson: add a second round where students revise the method and compare outcomes.
- Low-tech option: run with printed cards, sticky notes, and a board tally.

Extension Tasks

- Design a second version of the activity that tests the same knowledge issue in another AOK.
- Find a real-world example where the same knowledge problem matters outside the classroom.
- Write a 150-word TOK exhibition-style commentary using the activity as the object context.

Related Activities

- Loaded Headline Laboratory
- Metaphor Changes Policy
- Translation Loss Game